CSE (Cyber Security) B.Tech / B.E. • Discipline: COMPUTING & SECURITY

# CSE - Cyber Security

## 4-Year Comprehensive Career & Curriculum Action Plan

Defending digital sovereignty, enterprise cloud infrastructure, networks, and applications against global cyber threats.

Industry Context: Massive hiring demand in cybersecurity consultancies, banks, defence contractors, cloud providers, and Fortune 500 SOC teams.

### 1. TARGET INDUSTRY ROLES

Security Operations Center (SOC) Analyst • Penetration Tester / Ethical Hacker • Application Security (AppSec) Engineer • Cloud Security Architect • Incident Response Specialist

### 2. CORE TECHNICAL STACK

Core: Linux Internals & Bash, Computer Networks (TCP/IP, Wireshark), Python / C, Cryptography Basics
Specialization: Reverse Engineering (Ghidra, IDA), Binary Exploitation, Cloud Security (AWS IAM, GuardDuty), SIEM / Threat Hunting (Splunk)

### 3. MAJOR PORTFOLIO PROJECT PATH

Final Capstone Defense & submission of security tool / whitepaper to national cybersecurity conference

### 4. PLACEMENT & GATEWAY TRAJECTORY

Entry Tier: ¹6 LPA - ¹16 LPA
Mid Tier: ¹18 LPA - ¹36 LPA | Peak: ¹42 LPA - ¹80
Gateway: GATE CS / Specialized Cyber Forensics Entrance (IIT Kanpur (C3i Hub

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**MAN OF STUDY • Student Career Planning Platform**

Comprehensive 8-Semester Curriculum Architecture • Engineered for Day-1 Placement Success & Technical Depth

## 1. STRATEGIC CAREER & COMPENSATION OVERVIEW

Industry compensation tiers, career trajectory, and tier-wise hiring dynamics.

| ENTRY TIER (0-2 YRS) | MID TIER (3-6 YRS) | PEAK / GLOBAL TIER |
|---|---|---|
| ₹6 LPA - ₹16 LPA | ₹18 LPA - ₹36 LPA | ₹42 LPA - ₹80+ LPA (Consul |
| Campus Hiring & Day-1 Roles | Senior Engineer / Specialist | Staff Architect / Global Lead |

**Primary Industry Roles: Security Operations Center (SOC) Analyst • Penetration Tester / Ethical Hacker • Application Security (AppSec) Engineer • Cl Incident Response Specialist**

Corporate Viability & Scope: Massive hiring demand in cybersecurity consultancies, banks, defence contractors, cloud providers, and Fortune 500 SOC teams.

Government & PSU Opportunities: Primary recruitment channel for CERT-In, NTRO, Intelligence Bureau, DRDO SAG, Defence Cyber Agency, and banking PSUs (RBI,

**Apex / Tier-1 Campus Reality: Recruited directly by Big 4 cyber advisory, global investment banks (Goldman Sachs, Morgan Stanley cyber divisions), and cloud security providers.**

**Regional / Tier-3 Campus Reality: College curriculum rarely covers practical offensive security. Students from tier-3 who grind TryHackMe / HackTheBox (Top 10%), earn OSCP/CEH, and write CVE exploit writeups get hired directly into specialist cyber firms.**

**Key Off-Campus Placement Tactics:**

1. Earn rank in Top 5% on TryHackMe or HackTheBox; link public badge on resume and LinkedIn.
2. Publish responsible disclosure bug bounty reports or write-ups on HackerOne / Bugcrowd.
3. Contribute to open-source security tools (Python automation scripts, Burp Suite extensions).
4. Attain industry credentials such as CompTIA Security+, CEH, or prepare for OSCP.

## 2. CORE SKILLS & TECHNICAL STACK MATRIX

Structured competency progression from fundamental concepts to production tools.

Foundational Core: Linux Internals & Bash, Computer Networks (TCP/IP, Wireshark), Python / C, Cryptography Basics, Web Security (OWASP Top 10)

Advanced Specialization: Reverse Engineering (Ghidra, IDA), Binary Exploitation, Cloud Security (AWS IAM, GuardDuty), SIEM / Threat Hunting (Splunk), DevSecOps

Software, Tooling & DevOps: Kali Linux, Burp Suite, Wireshark, Metasploit, Nmap, Splunk, Ghidra, Docker, TryHackMe / HackTheBox

Higher Studies Gateway (GATE CS / Specialized Cyber Forensics Entrance): Top Institutes: IIT Kanpur (C3i Hub), IIT Madras (Cyber Security), NFSU Gandhinagar (Forensics & Cyber Security), IISc Bengaluru | M.Tech Tracks: Cyber Security, Information Security & Trustworthy Systems, Cyber Physical Systems Security | Scope Abroad: Massive demand across US (Carnegie Mellon, Johns Hopkins, NYU Tandon), UK, and Netherlands.

## 3. 4-YEAR SEMESTER-BY-SEMESTER ACTION PLAN

Exhaustive 8-semester roadmap covering core subjects, skills, tools, projects, and milestones.

## YEAR 1: LINUX FLUENCY, NETWORKING & C PROGRAMMING — Low-level computer understanding, network protocols, and

### SEMESTER 1: Semester 1: Linux Systems & C Programming

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|---|---|
| **Core Subjects:** | Engineering Mathematics I, Problem Solving in C, Digital Electronics, Basic Electrical |
| **Industry Skills:** | Linux CLI, Shell scripting (Bash), File permissions, Memory pointers in C, Git |
| **Tools & Platforms:** | Ubuntu / Debian CLI, GCC, Bash, GitHub |
| **Build / Project:** | Custom Linux Shell with command piping and process execution; CLI Port Scanner in C |
| **Career Milestone:** | Establish Linux as daily driver OS; earn OverTheWire Bandit wargame completion. |
| **Placement Prep:** | Build fundamental problem-solving habits in C/C++. |

### SEMESTER 2: Semester 2: Computer Networks & Foundational Python

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|---|---|
| **Core Subjects:** | Engineering Mathematics II (Discrete Math), Computer Networks, Python Programming, Object-Oriented Design |
| **Industry Skills:** | OSI model, Packet sniffing, Wireshark analysis, Subnetting, Socket programming in Python |
| **Tools & Platforms:** | Wireshark, Python 3, Packet Tracer, TryHackMe (Pre-Security path) |
| **Build / Project:** | Custom Network Packet Sniffer & Protocol Analyzer in Python; Encrypted Peer-to-Peer Chat Application using Socket API and AES |
| **Career Milestone:** | Reach Top 10% ranking on TryHackMe; solve 50+ basic cybersecurity labs. |
| **Placement Prep:** | Master TCP/IP handshakes, routing protocols, and subnet calculations. |

## YEAR 2: WEB SECURITY, CRYPTOGRAPHY & PENETRATION TESTING — Hands-on offensive testing, OWASP

### SEMESTER 3: Semester 3: Web Application Security & OWASP Top 10

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|---|---|
| **Core Subjects:** | Database Management Systems, Design & Analysis of Algorithms, Applied Cryptography, Operating Systems |
| **Industry Skills:** | SQL Injection, XSS, CSRF, IDOR, SSRF, Symmetric/Asymmetric encryption, RSA, Hashing |
| **Tools & Platforms:** | Burp Suite Community, OWASP Juice Shop, DVWA, PostgreSQL |
| **Build / Project:** | Comprehensive Vulnerability Assessment Report on Intentionally Vulnerable Web App; Cryptographic File Vault CLI with AES-256 and RSA key exchange |
| **Career Milestone:** | Achieve PortSwigger Web Security Academy Practitioner status. |
| **Placement Prep:** | Prepare for CompTIA Security+ or Certified Ethical Hacker (CEH) certification. |

### SEMESTER 4: Semester 4: Network Penetration Testing & System Hardening

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|---|---|
| **Core Subjects:** | Network Security & Firewalls, Theory of Computation, Software Engineering, System Administration |
| **Industry Skills:** | Port scanning, Service enumeration, Metasploit, Privilege escalation (Linux & Windows), IPTables, Snort IDS |
| **Tools & Platforms:** | Kali Linux, Nmap, Metasploit, Snort, HackTheBox |
| **Build / Project:** | Automated Vulnerability Scanner & Network Auditor in Python; Configuring and Testing a Snort Intrusion Detection System against simulated attacks |
| **Career Milestone:** | Submit responsible vulnerability disclosure (bug bounty program) or earn HackTheBox Pro Hacker rank. |
| **Placement Prep:** | Study for GATE CS networking and OS sections; attend collegiate CTF competitions. |

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## YEAR 3: SOC OPERATIONS, CLOUD SECURITY & DEVSECOPS — Defensive engineering, threat intelligence, cloud security

### SEMESTER 5: Semester 5: Cloud Security & Defensive Operations (SOC)

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| **Core Subjects:** | Cloud Infrastructure Security, Information Security Management (ISO 27001), Cyber Forensics & Incident Response, Elective: Malware Analysis |
| **Industry Skills:** | AWS IAM policy least-privilege, SIEM log analysis, Splunk query language, PCAP forensic analysis, Volatility memory analysis |
| **Tools & Platforms:** | Splunk Free, AWS IAM / GuardDuty, Volatility, Ghidra, Docker |
| **Build / Project:** | Cloud Security Compliance Auditor for AWS S3 and IAM roles using Python and Boto3; Memory Forensic Investigation Report of a simulated ransomware attack |
| **Career Milestone:** | Land a corporate cybersecurity summer internship (SOC Analyst or Security Consultant intern). |
| **Placement Prep:** | Intensive interview prep for corporate security consulting firms (Big 4, CrowdStrike, Palo Alto Networks). |

### SEMESTER 6: Semester 6: DevSecOps, Binary Analysis & Threat Hunting

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|---|---|
| **Core Subjects:** | DevSecOps & Secure Coding, Malware Analysis & Reverse Engineering, Elective: IoT Security, Mini-Project |
| **Industry Skills:** | SAST / DAST integration in CI/CD, Dependency scanning, Buffer overflows, Disassembly with Ghidra |
| **Tools & Platforms:** | SonarQube, Snyk, GitHub Actions, Ghidra, Prowler |
| **Build / Project:** | Automated DevSecOps Pipeline with SAST, container vulnerability scanning, and compliance enforcement; Reverse Engineering a custom crackme binary and authoring a detailed report |
| **Career Milestone:** | Complete corporate cybersecurity internship; aim for Pre-Placement Offer (PPO). |
| **Placement Prep:** | Begin studying materials for OSCP (Offensive Security Certified Professional). |

## YEAR 4: CAPSTONE, CERTIFICATIONS & DEFENSE RECRUITMENT — Enterprise security defense, campus placement

### SEMESTER 7: Semester 7: Campus Placement Drives & Capstone Stage 1

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|---|---|
| **Core Subjects:** | Cyber Laws, Ethics & Cyber War, Major Project Phase 1, Open Elective |
| **Industry Skills:** | Threat modeling (STRIDE), Red team vs Blue team adversary simulation, Executive communication |
| **Tools & Platforms:** | MITRE ATT&CK Framework, Splunk, AWS, LinkedIn |
| **Build / Project:** | Production Capstone: Automated Threat Hunting Platform mapped to MITRE ATT&CK Matrix |
| **Career Milestone:** | Secure Day-1 cybersecurity campus placement or off-campus cyber firm offer. |
| **Placement Prep:** | Appear for DRDO RAC / NTRO screening examinations if interested in national defense cadres. |

### SEMESTER 8: Semester 8: Final Defense & Professional Onboarding

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|---|---|
| **Core Subjects:** | Major Project Phase 2, Industry Internship / Dissertation |
| **Industry Skills:** | Incident handling runbooks, Security architecture reviews, Production infrastructure auditing |
| **Tools & Platforms:** | Jira, Confluence, GitLab |
| **Build / Project:** | Final Capstone Defense & submission of security tool / whitepaper to national cybersecurity conference |
| **Career Milestone:** | Transition into full-time corporate AppSec / SOC Analyst or government cyber posting. |
| **Placement Prep:** | Complete corporate onboarding with active industry credentials. |

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## 4. STRATEGIC TIPS FOR CRACKING HIGH-GROWTH PLACEMENTS

Actionable advice from industry leaders, alumni mentors, and technical recruiters.

1. Active participation on platforms like TryHackMe, HackTheBox, and CTFtime speaks louder than academic GPA in technical security interviews.
2. Learn both offensive tactics and defensive engineering; modern high-paying security roles demand "Purple Team" thinking.
3. Never run aggressive scans or tests against targets without explicit, written authorization; always operate in strictly controlled lab environments.

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